CALCULAS

Date:

Function :-

A function is a relation between set of Inputs (Domain) and Output (Range). Each input is related to exactly one output.

$$f : X \to Y$$

Where $y = f(x)$, $\forall x \in X$ and $y \in Y$.

Example: If function $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = 3x + 5$, then find.

i) $f(2)$ ii) $f(-3)$ iii) $f(2/3)$

Solve

i) $f(2) = 3(2) + 5 = 11$

ii) $f(-3) = 3(-3) = -4$

iii) $f\left(\frac{2}{3}\right) = 3\left(\frac{2}{3}\right) + 5 = 7$

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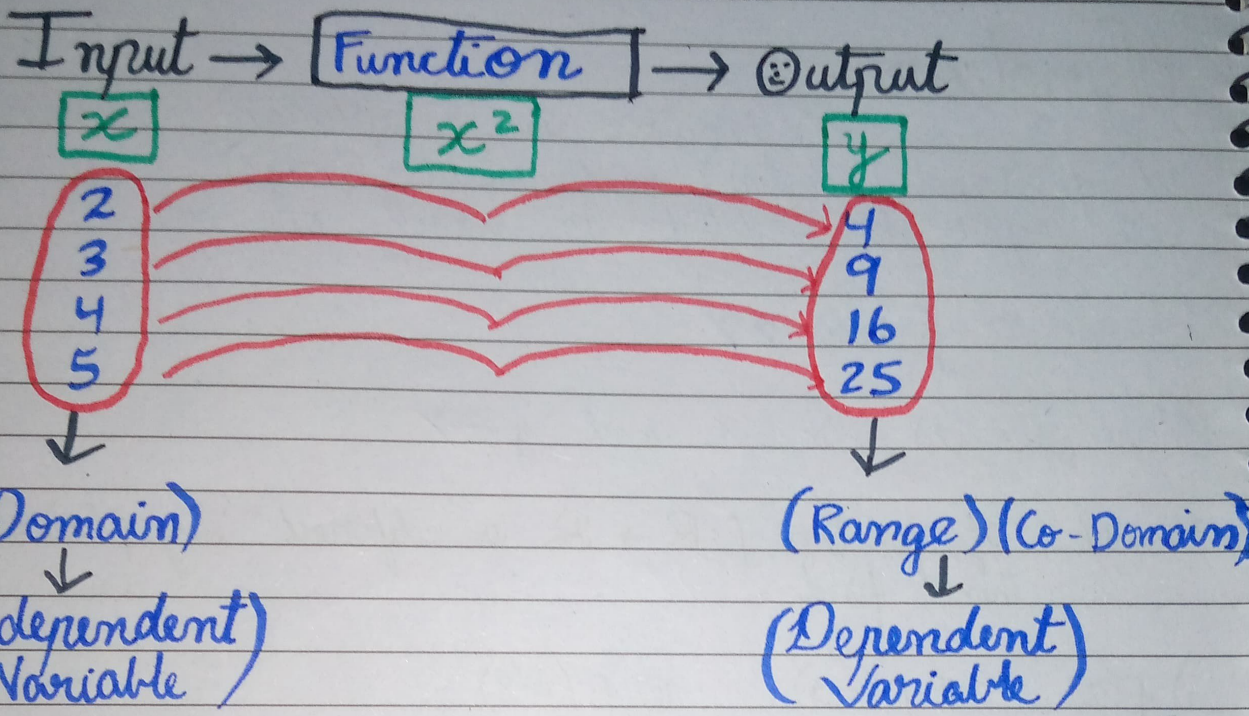
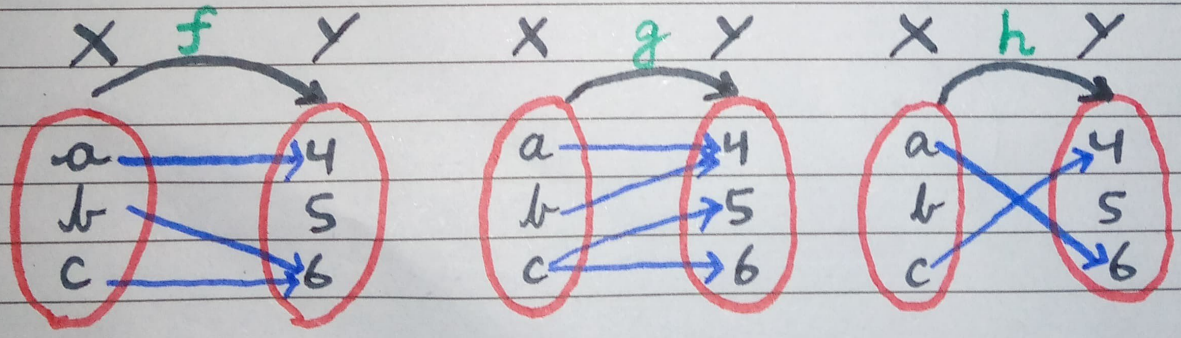
Date: _____



Example: Let $X = \{a, b, c\}$ and $Y = \{4, 5, 6\}$. State whether or not the relations indicated by the following figures are function from X to Y.



f is a function, because each element of X has a unique image.
g is not a function, because the element c has 2 images in Y.
h is not a function, because the element B has no image in Y.

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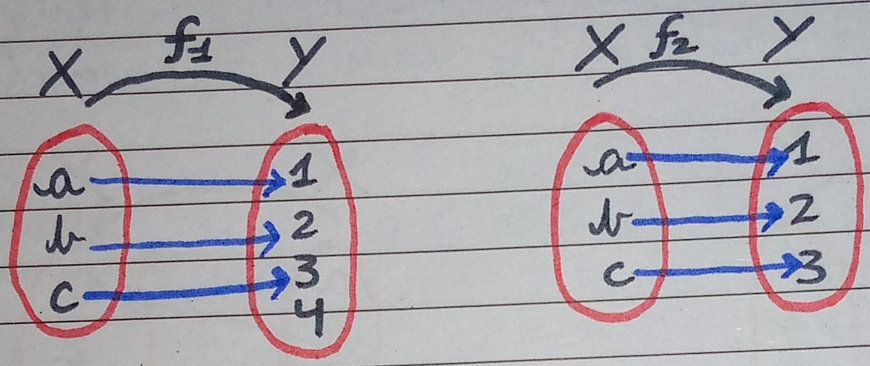
One to one & Onto Functions.

Date:

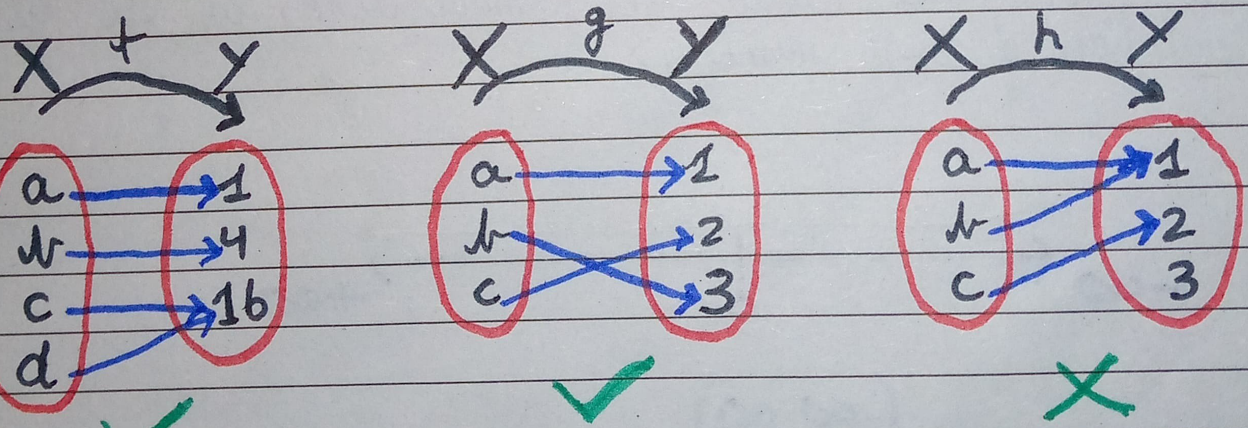
i) One-to-one Function (Injective) :-

A function $f: X \to Y$ is one-to-one (injective) if distinct elements of set X have distinct images in set Y. And X can also have some left out elements.



ii) Onto Function (Surjective) :-

A function is called surjective if every element in the co-domain has atleast one pre-image in the domain.



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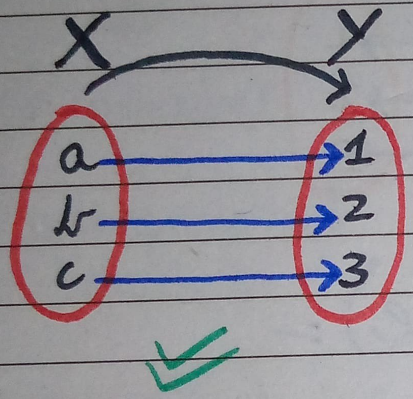
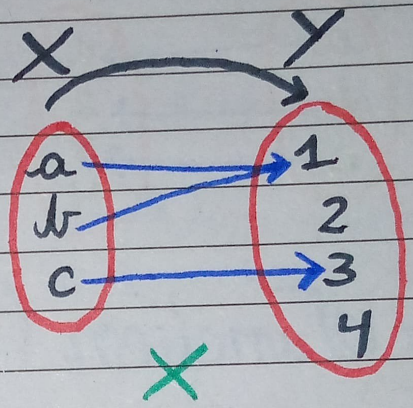
Date:

iii) One to One & Onto (Bijective) :-

A function is called bijective if it is both injective and surjective. That means:

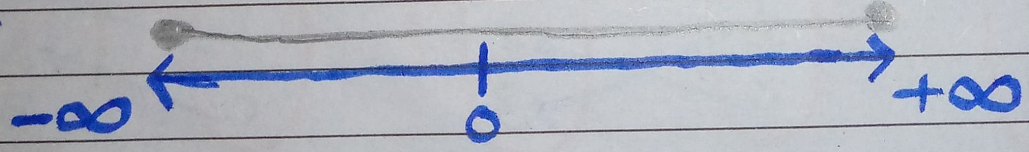
- Every input has a unique output (injective), AND
- Every output is matched by some input (surjective).





What is the Real Line?

The real line (also called the number line) is a visual representation of real numbers.



$(-\infty, \infty)$

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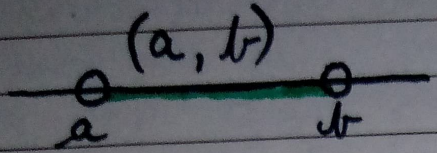
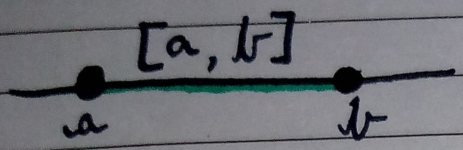
Intervals Notation

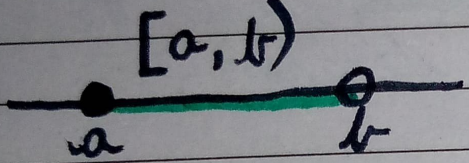
Date:



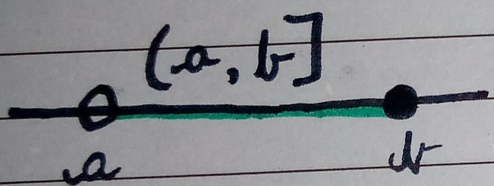
$a < x < b$, Open interval



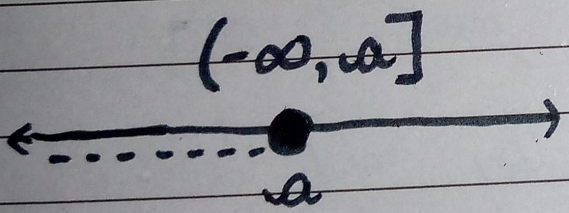
$a \leq x \leq b$, Closed interval



$a \leq x < b$, Semi Open interval



$a < x \leq b$, Semi Open interval



$x \leq a$, Infinite interval

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⋆ What is Domain?

Date:

The domain of a function is the set of all input values (x-values) for which the function is defined (i.e works without errors like division by 0 or square root of a given number).

⋆ What is Range?

The range (co-domain) is the set of all possible outcomes (output values) that the function can produce from its domain.

Key this rules in mind:

Rational functions: Denominator $\neq 0$ → Exclude x-values that make denominator 0.

Square roots ($\sqrt{}$): Expression under $\sqrt{}$ must be ≥ 0.

Logarithmic functions: Argument inside log must be > 0.

Piecewise functions: Watch domain of each piece carefully

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$f(x) = \sqrt{x-5}$

$y = \frac{x+1}{x-4}$

$y = x^2$

Date:

Q Find the domain and range of the following.

$f = \frac{1}{x}$

$y = \frac{1}{x-3}$

$y = \frac{1}{x^2-9}$

$y = \frac{1}{x^2+9}$

$f(x) = \sqrt{x}$

$f(x) = \sqrt{x+3}$

$f(x) = \frac{1}{\sqrt{25-x}}$

$f(x) = \sqrt{x^2-4}$

$f(x) = \sqrt{2x-1}$

$f(x) = \sqrt{5-x}$

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Composite Function

Date:

A composite function is when you apply one function inside another. You are literally feeding the output of one function into another function. We write it as:

$$(f \circ g)(x) = f(g(x))$$

This means: First apply g to x, then apply f to the result of $g(x)$.

- $(f \circ g)(x) \neq (g \circ f)(x)$
- Always start from the inside (first compute $g(x)$, then $f(g(x))$.

Example: If $f(x) = x^2$ and $g(x) = x - 3$. Find the composite function $(f \circ g)(x) = f(g(x))$.

$f(x) = x^2$

$f(x-3) = (x-3)^2$

$f(g(x)) = x^2 - 6x + 9$

$\boxed{f \circ g = x^2 - 6x + 9}$

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Steps to find Inverse :-

Date:

Suppose $f(x) = y$, follow these steps.

1) Replace $f(x)$ with y:

$$y = f(x)$$

2) Swap x and y:

$$x = f(y)$$

3) Solve for y.

4) Replace y with $f^{-1}(x)$.

Example: Lets find the inverse of: $f(x) = 3x + 2$

$$y = 3x + 2$$
$$x = 3y + 2$$
$$3y = x - 2$$
$$y = \frac{x-2}{3}$$
$$\boxed{f^{-1}(x) = \frac{x-2}{3}}$$

Check:- $f\left(f^{-1}(x)\right) = 3\left(\frac{x-2}{3}\right) + 2 = x$

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Inverse Function

Date:

If a function f maps $x \to y$, then its inverse function f^{-1} maps $y \to x$.

Let's say we have a function :

$$f(x) = y$$

The inverse function does the opposite.

$$f^{-1}(y) = x$$

Formal Definition :-

A function f has inverse if :

$$f(f^{-1}(x)) = x \quad ; \quad f^{-1}(f(x)) = x$$

Condition for Inverse Function to Exist :-

Not every function has an inverse.

For a function to have an inverse, it must be :

1) Injective (one to one)
2) Surjective (onto)

for inverse to exist as a function the original function must be bijective (both one to one and onto)

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Date:

Q: If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find

a) $f \circ f$ b) $g \circ g$ c) $f \circ g$ d) $g \circ f$

a) solve

$$f \circ f = f(f(x))$$

$$f(\sqrt{x}) = \sqrt{\sqrt{x}}$$

$$= \sqrt[4]{x}$$

b) solve

$$g \circ g = g(g(x))$$

$$= g(\sqrt{2-x})$$

$$= \sqrt{2-\sqrt{2-x}}$$

c) solve

$$f \circ g = f(g(x))$$

$$f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}}$$

$$= \sqrt[4]{2-x}$$

d) solve

$$g \circ f = g(f(x))$$

$$g(\sqrt{x}) = \sqrt{2-\sqrt{x}}$$

Q: $f(x) = \frac{1}{x+2}$, $g(x) = x+1$. Find

a) $f \circ f$ b) $g \circ g$ c) $f \circ g$ d) $g \circ f$

a) solve

$$f(f(x)) = \frac{1}{\left(\frac{1}{x+2}\right)+2}$$

b) solve

$$g(g(x)) = (x+1)+1$$

$$= x+1+1$$

$$= x+2$$

c) solve

$$f(g(x)) = \frac{1}{(x+1)+2}$$

$$= \frac{1}{x+3}$$

d) solve

$$g(f(x)) = \left(\frac{1}{x+2}\right)+1$$

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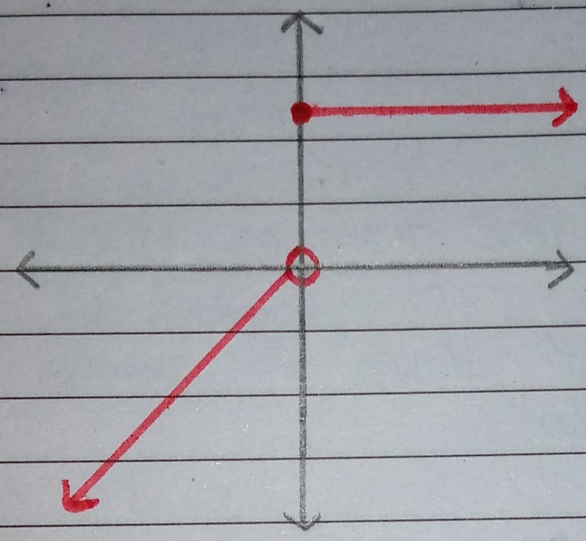
Date:_____

$$f(x) = \begin{cases} x, & x < 0 \\ 5, & x \geq 0 \end{cases}$$



Piecewise Functions :- Date:

A piecewise function is a function that is defined using different formulas (or expressions) for different parts (or intervals) of the domain.

Formal Definition:

$$f(x) = \begin{cases} f_1(x), & \text{if } x \in A_1 \\ f_2(x), & \text{if } x \in A_2 \\ \vdots & \\ f_n(x), & \text{if } x \in A_n \end{cases}$$

Where:

- Each $f_1(x)$ is a different expression.
- Each A_1 is a different interval / domain condition.

Example :-

$$f(x) = \begin{cases} x+2, & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$$

This means:

- If $x = -3$, use the 1st expression $\rightarrow f(-3) = -3+2 = -1$
- If $x = 2$, use the second $\rightarrow f(2) = 2^2 = 4$

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Absolute Value Function

Date:

Definition :-

The absolute value of a number tells you how far it is from 0 on the number line.

Symbol :-

$|x|$

Definition (Piecewise Form) :-

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

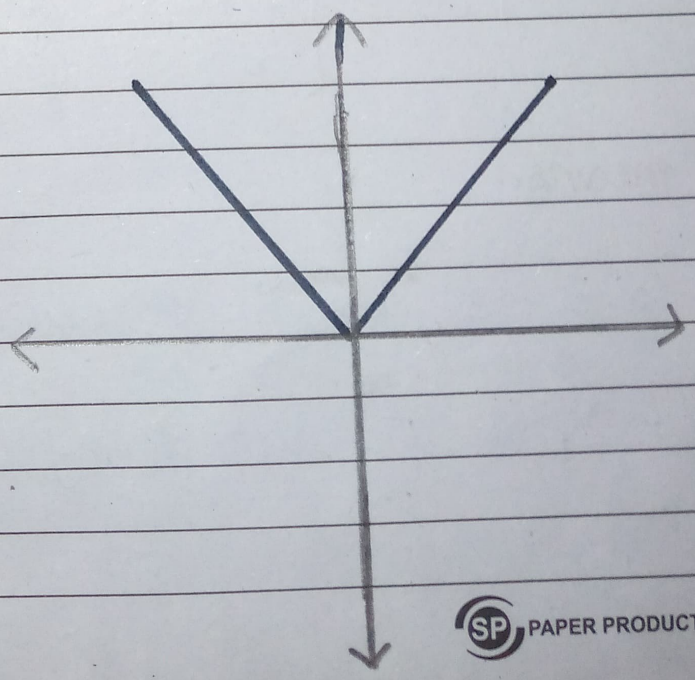
Example :-

$|5| = 5$; $|-5| = 5$, $|0| = 0$

* Distance is always positive or zero never negative.

Domain $(-\infty, \infty)$

Range $[0, \infty)$



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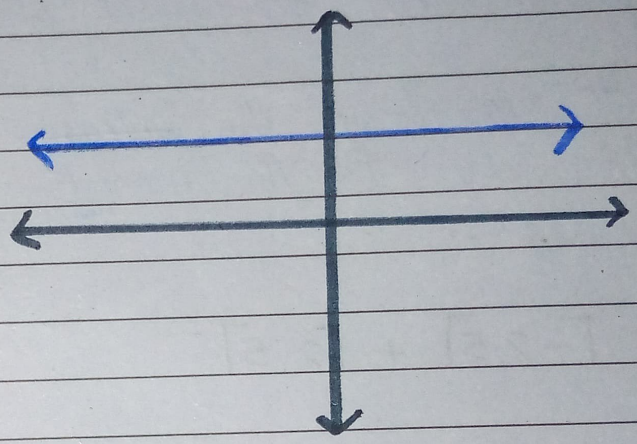
Basic Graph Shapes Of Common Functions :-

Date:

These are the 8 essentials graphs you must memorize and understand perfectly. Each one builds your intution for bigger topics.

1) Constant Function :-

$$f(x) = c$$



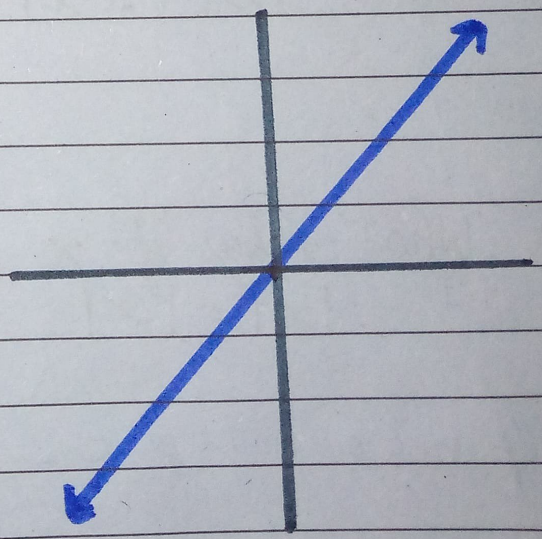
Where c is a constant number.

Shape : A horizontal line.

Graph Info : Domain : $(-\infty, \infty)$
Range : (c)

2) Indentity Function :-

$$f(x) = x$$



Shape : A diagonal line passing through the origin at 45°.

Domain : $(-\infty, \infty)$
Range : $(-\infty, \infty)$
Slope = 1
passes through origin $(0, 0)$

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Ceiling and Floor functions both are piecewise functions.

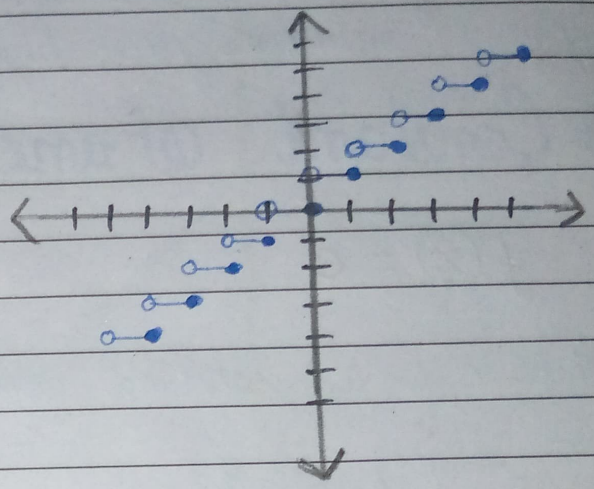
Ceiling Function :-

Date: _____

The ceiling function of a number x, written as:

$$f(x) = \lceil x \rceil$$

- It is the smallest integer greater than or equal to x.
- It rounds up the ~~integer~~ ~~to~~ number to the nearest integer.



x	$\lfloor x \rfloor$	$\lceil x \rceil$
4.6	4	5
3.7	3	4
3.3	3	4
2.5	2	3
1.9	1	2
0	0	0
-2	2	2
-1.7	-2	-1
-2.5	-3	-2
-3.9	-4	-3
-4.1	-5	-4
-7.7	-8	-7

Q:1) $\lceil -2.5 \rceil + \lceil 2.5 \rceil$

$\lceil -2.5 \rceil = -2$, $\lceil 2.5 \rceil = 3$

$-2 + 3 = 1$

Q:2 Solve $\lceil x \rceil + \lceil -x \rceil = 1$

$x =$ Decimals

Q:3 What values of x satisfy $\lceil x \rceil = \lfloor x \rfloor$

If and only if $x \in \mathbb{Z}$

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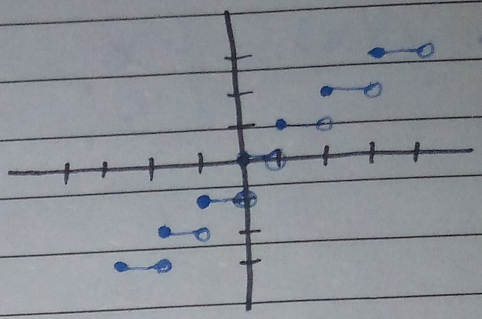
Floor Function (Greatest Integer Function) :-

Date: _____

The floor function, written as:

$$f(x) = \lfloor x \rfloor$$



means:

"The greatest integer less than or equal to x."

In simple words: Round down the number to the nearest whole number.

Properties of $\lfloor x \rfloor$:-

1) $x - 1 < \lfloor x \rfloor \leq x$
2) if $x \in \mathbb{Z}$, then $\lfloor x \rfloor = x$.
3) Floor function is not continuous at integers (has a jump)

Q1 Find $\lfloor 5.3 \rfloor + \lfloor -2.8 \rfloor$

$\lfloor 5.3 \rfloor = 5$, $\lfloor -2.8 \rfloor = -3$

$5 + (-3) = 2$

Q:2 If $\lfloor x \rfloor + \lfloor -x \rfloor = -1$, find the possible values of x.

All non-integer real numbers (decimals).

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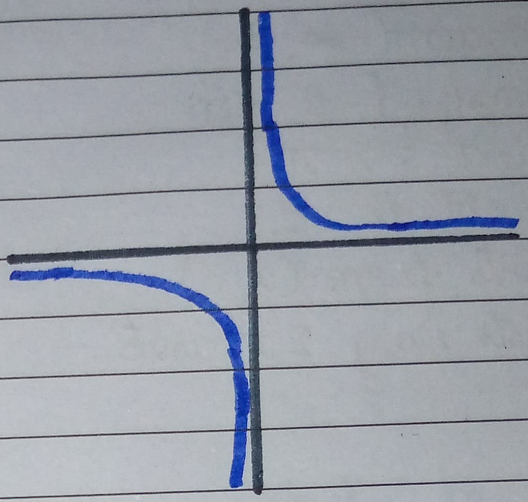
Date:

7) Reciprocal Function :-

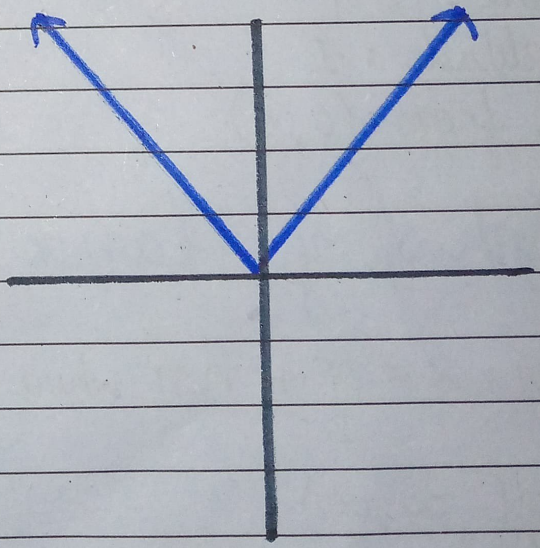
$$f(x) = \frac{1}{x}$$



- Two branches in 1st and 3rd quadrants.
- Hyperbola Shape
- Domain : $(-\infty, 0) \cup (0, \infty)$
- Range : $(-\infty, 0) \cup (0, \infty)$
- Asymptotes : $x = 0, y = 0$
- Never touches x or y axis.

8) Absolute Value Function :-

$$f(x) = |x|$$



- A V-shaped graph with vertex at origin.
- Domain : $[-\infty, \infty)$
- Range : $[0, \infty)$
- Symmetric about y-axis.
- Minimum at $x = 0$

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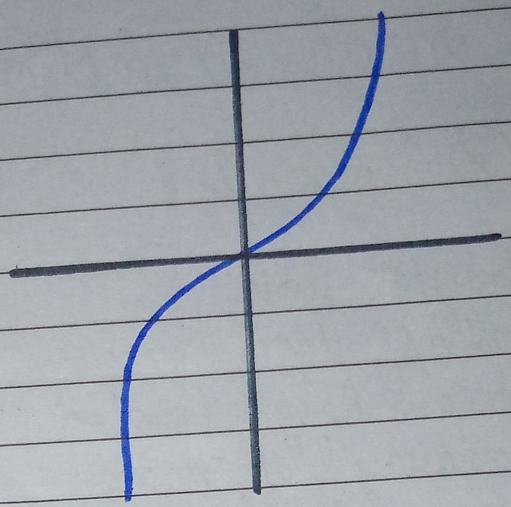
Date:

5) Cubic Function :-

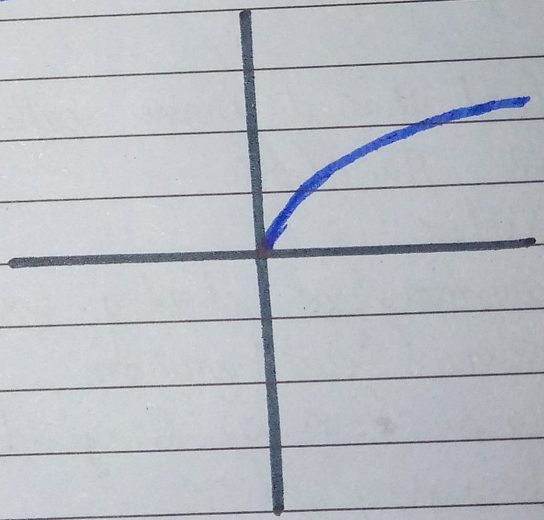
$$f(x) = x^3$$



- An S shaped curve that increase throughout.
- Pass through origin
- Symmetric about origin (odd Function)
- Domain & Range: $(-\infty, \infty)$
- Increasing everywhere

6) Square Root Function :-

$$f(x) = \sqrt{x}$$



- A curve starting from origin and increasing slowly.
- Domain: $[0, \infty)$
- Range : $[0, \infty)$
- Only in first quadrant.
- Not defined for negative values of x.

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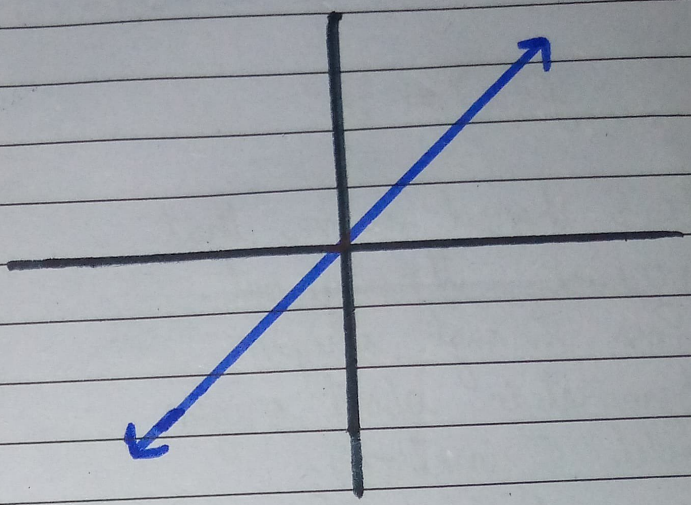
Date:

3) Linear Function :-

$f(x) = mx + b$

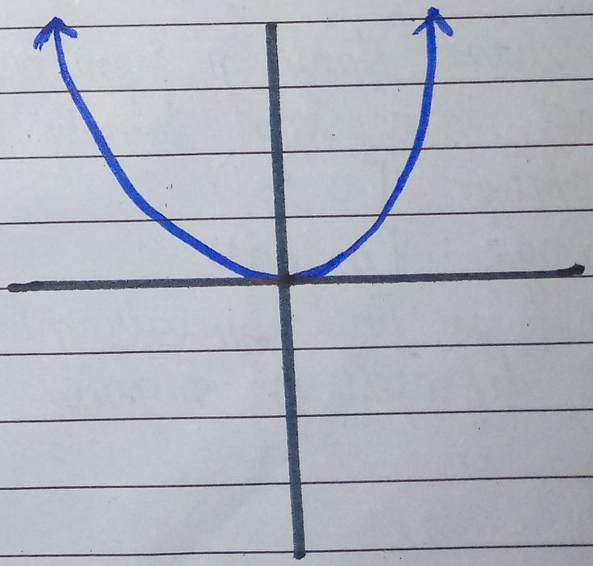
- Straight lines
- Constant slopes
- Domain & Range: $(-\infty, \infty)$
- Increasing if $m > 0$, decreasing if $m < 0$.



4) Quadratic Function :-

$f(x) = x^2$

- A U-shaped curve called a parabola.
- Vertex : $(0, 0)$
- Symmetric about y-axis means (Even Function)
- Domain: $(-\infty, \infty)$, Range : $[0, \infty)$
- Changing slope



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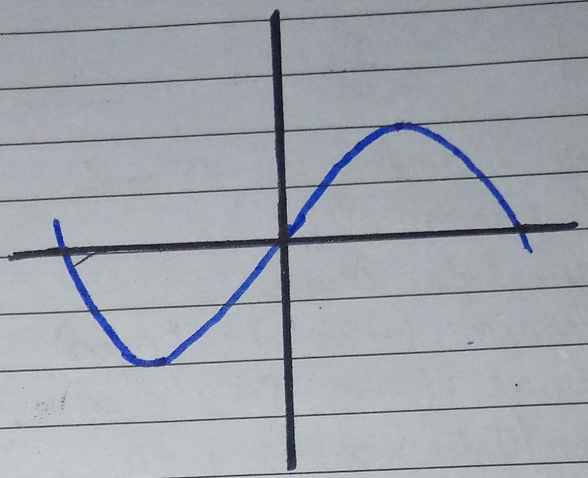
Trignometric Graphs :-

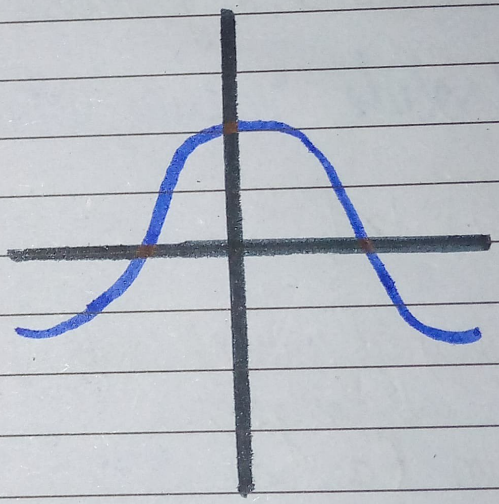
Date:

1) Graph of $y = \sin x$



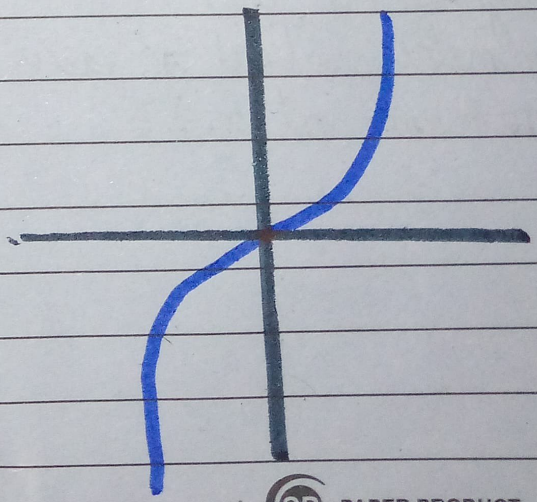
- Domain: $(-\infty, \infty)$
- Range: $[-1, 1]$
- Period : 2π
- Amplitude : 1
- Starts from: $(0, 0)$
- Repeats every 2π units.

2) Graph of $y = \cos x$



- Domain : $(-\infty, \infty)$
- Range : $[-1, 1]$
- Period : 2π
- Amplitude : 1
- Starts from : $(0, 1)$

3) Graph of $y = \tan x$



- Domain : $x \neq \frac{\pi}{2} + n\pi$, where $n \in \mathbb{Z}$.
- Range: $(-\infty, \infty)$
- Period : π
- No amplitude (it goes to ∞)
- Steep S-curve shape.
- Has vertical asymptotes where cosine = 0.

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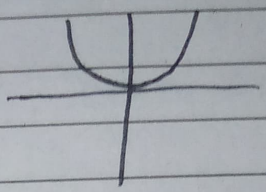
Graph Transformations (Shifts, Reflect, Stretch, Compress)

Date:

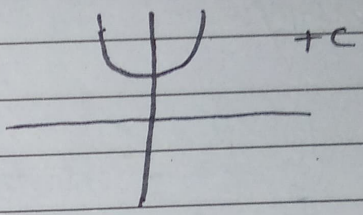
These allow you to transform basic graphs to create new ones without plotting points from scratch.

1) Vertical Shifts :-

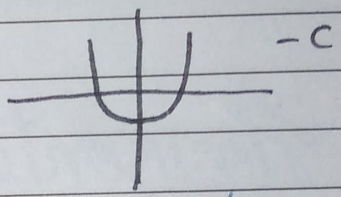


Rule:

If $f(x)$ your original function, then



- $f(x) + c \rightarrow$ shift up by c units.
- $f(x) - c \rightarrow$ shift down by c units.

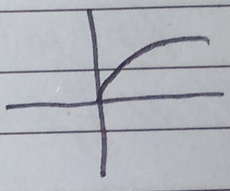


Example:

- $f(x) = x^2 \rightarrow$ U-shaped parabola
- $f(x) = x^2 + 3 \rightarrow$ same parabola, shifted up 3 units.

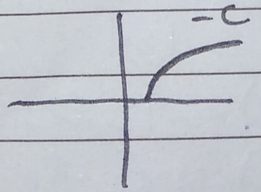
2) Horizontal Shifts :-

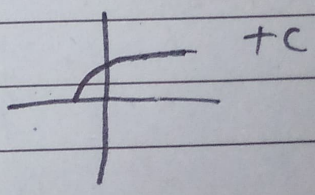
Rule:
- $f(x - c) \rightarrow$ shift right by c units.
- $f(x + c) \rightarrow$ shift left by c units.



Important: Horizontal shifts are opposite in sign.

Example:





- $f(x) = \sqrt{x}$
- ~~$f(x) =$~~ $f(x-2) = \sqrt{x-2} \rightarrow$ shift right by 2 units.

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Date:

3) Vertical Stretch & Compression :-

- $a \cdot f(x)$, where $a > 1$: Stretch Vertically
- $0 < a < 1$: Compress Vertically

Example :-

$f(x) = x^2$
$2x^2$: stretch taller
$\frac{1}{2}x^2$: flatter parabola

4) Horizontal Stretch & Compression :-

- $f(ax)$, where $a > 1$: Compress Vertically
- $0 < a < 1$: Stretch horizontally.

Horizontal changes are inversely related to the coefficient.

Example :-

- $f(x) = |x|$
  $f(2x) = |2x|$: Compressed (narrow V)
  $f(\frac{1}{2}x) = |\frac{1}{2}x|$: wider V

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Example 2: Function not Defined at a point

Let: $f(x) = \frac{x^2 - 1}{x - 1}$

Now try plugging $x = 1$:

$f(1) = \frac{1^2 - 1}{1 - 1} = \frac{0}{0}$ × Undefined

But we can still find:

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$$

Lets simplify:

$$\frac{x^2 - 1}{x - 1} = \frac{(x-1)(x+1)}{(x-1)} = x + 1$$

So the limit becomes:

$$\lim_{x \to 1} (x + 1) = 2$$

* Even though the function is undefined at 1, the limit exists.

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What is a Limit :-

Date: _____

The limit of a function describes the value that the function approaches as the input gets closer and closer to a particular number.

In simple words :

"What value is $f(x)$ getting closer to, as x gets close to a specific value ?"

Example :-

$$f(x) = x^2$$

What is :

$$\lim_{x \to 2} f(x) = ?$$

We just plug in :

$$f(2) = 2^2 = 4$$

$$\lim_{x \to 2} x^2 = 4$$

Note : The value of the limit depends on the values that $f(x)$ is getting close to around $x = 2$, not necessarily at $x = 2$.

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Always follow this order while transforming a graph:

1) Horizontal shifts
2) Horizontal stretch/Compression
3) Reflections
4) Vertical Stretch/Compression
5) Vertical shifts.

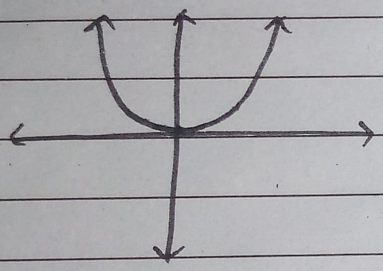
Date: _____

5) Reflections

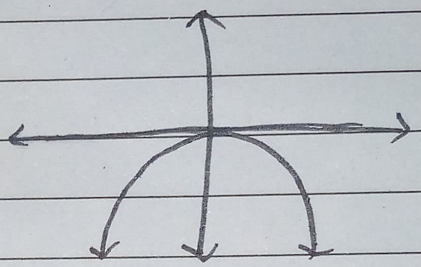
- $-f(x) = -y$: Reflects over x-axis (flip vertically).

Examples:-

$f(x) = x^2$



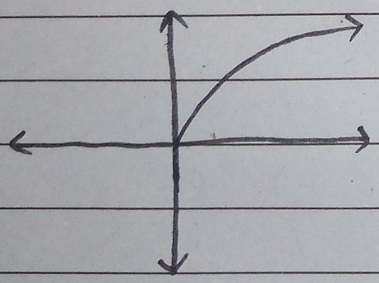
$g(x) = -x^2$



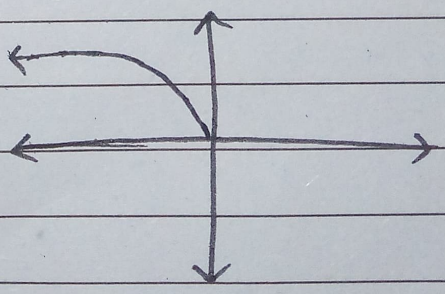
- $f(-x)$: Reflect over y-axis (flip horizontally).

Example:-

$f(x) = \sqrt{x}$



$g(x) = \sqrt{-x}$



Reflected left (but may become undefined).

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3) Limit Laws :-

Date:

You can apply these rules only when individual limit exist.

3.1) Sum Rule :-

$$\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} [f(x)] + \lim_{x \to a} [g(x)]$$

3.2) Difference Rule :-

$$\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} [f(x)] - \lim_{x \to a} [g(x)]$$

3.3) Product Rule :-

$$\lim_{x \to a} [f(x) \cdot g(x)] = \lim_{x \to a} [f(x)] \cdot \lim_{x \to a} [g(x)]$$

3.4) Quotient Rule :-

$$\lim_{x \to a} \left[\frac{f(x)}{g(x)}\right] = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}, \text{ if } \lim_{x \to a} g(x) \neq 0$$

3.5) Power Rule :-

$$\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x)\right]^n$$

3.6) Constant Multiple Rule :-

$$\lim_{x \to a} [c \cdot f(x)] = c \cdot \lim_{x \to a} [f(x)]$$

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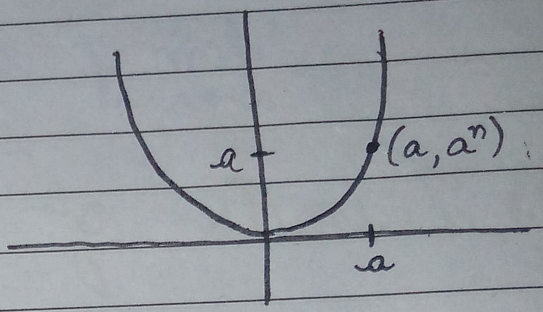
2.3) Limit of Power Functions :-

- $f(x^n) = x^n$
- $\lim_{x \to a} f(x^n) = a^n$

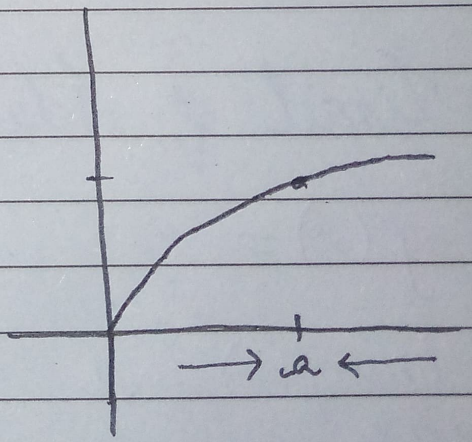


Examples:

$\lim_{x \to 3} x^2 = 9$, $\lim_{x \to -2} x^3 = -8$

2.4) Limit of Root Functions :-

- $\lim_{x \to a} \sqrt{x} = \sqrt{a}$ (only when $a \geq 0$)
- $\lim_{x \to a} \sqrt[n]{x} = \sqrt[n]{a}$



Examples:

$\lim_{x \to 4} \sqrt{x} = \sqrt{4} = 2$

$\lim_{x \to 0} \sqrt{x} = 0$

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1) Limit Definition :-

Date:

If the values of $f(x)$ approach a number L as x approaches a value, we write

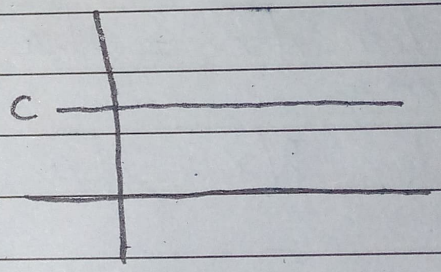
$$\lim_{x \to a} f(x) = L$$

2) Limits of Basic Functions :-

We will now find limits for some common functions using intution and graphs.

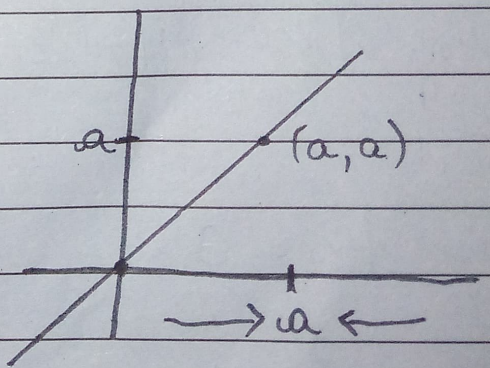
2.1) Limit of Constant Function :-

- $f(x) = c$
- $\lim_{x \to a} f(x) = c$



2.2) Limit of Identity Function :-

- $f(x) = x$
- $\lim_{x \to a} f(x) = a$



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4) Limits of Polynomial And Rational Functions :-

Date:

4.1) Polynomial Functions :-

- For any polynomial :

$$\lim_{x \to a} P(x) = P(a)$$

Example: $f(x) = 3x^2 - x + 4$, x tends to 2.

$$\lim_{x \to 2} f(x) = 3(2)^2 - 2 + 4$$

$$= 12 - 2 + 4$$

$$= 14$$

4.2) Rational Functions :-

Let $R(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials.

Case 1 : Plugging values gives a valid number.

- $Q(x) \neq 0$

Example: $f(x) = \frac{x^2 - 1}{x - 1}$

$$\lim_{x \to 0} f(x) = \frac{x^2 - 1}{x - 1} = \frac{0^2 - 1}{0 - 1} = \boxed{1}$$

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Case 2: plugging gives $\frac{0}{0}$

- Use factoring, conjugate, simplification.

Example: $\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$

$\Rightarrow \lim_{x \to 1} \frac{(x-1)(x+1)}{(x-1)}$

$\Rightarrow 1 + 1 = 2$

Case 3: Plugging gives $\frac{k}{0}$

- Means infinite limit of DNE. Analyze left/right behaviour

Example: $\lim_{x \to 1} \frac{x^2 - x - 2}{x^2 - 2x + 1}$

$\Rightarrow \lim_{x \to 1} \frac{(x-1)(x+2)}{(x-1)(x-1)} = \lim_{x \to 1} \frac{x+2}{x-1}$, $q(1) = 1 - 1 = 0$, still 0

$\Rightarrow \lim_{x \to 1} \frac{x+2}{x-1} = \frac{1+2}{1-1} = \frac{3}{0} = \infty$

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Left-Hand And Right-Hand Limits :-

Date:

A limit tells us what value a function approaches as x gets closer to a number.
But... sometimes from left side and from the right side, the function might approach different values. In that case, we consider:

- Left-hand limit (LHL):

$$\lim_{x \to a^-} f(x)$$

What $f(x)$ approaches as x approaches a from the left $(x < a)$.

- Right-hand limit (RHL):

$$\lim_{x \to a^+} f(x)$$

What $f(x)$ approaches as x approaches a from the right $(x > a)$.

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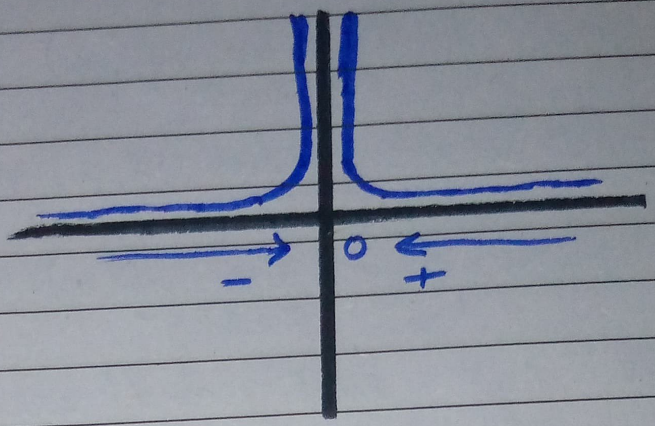
Date:

- Example:

$$f(x) = \frac{1}{x^2}$$



At $x = 0$:

- $\lim_{x \to 0^-} f(x) = +\infty$
- $\lim_{x \to 0^+} f(x) = +\infty$

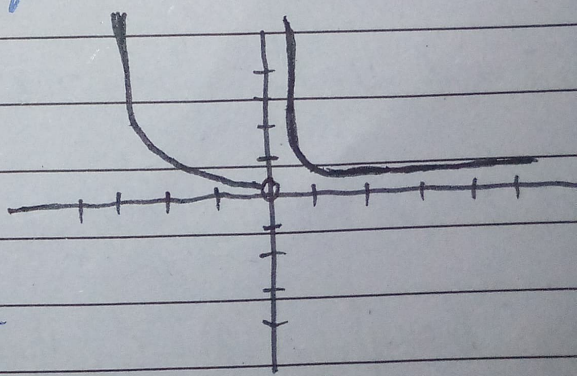
limit does not exist at $x = 0$. (it diverges to infinity).

Case 4: One side has a finite limit, other side goes to $\pm\infty$.

◆ Condition:

- $\lim_{x \to a^-} f(x) = \text{finite}$, $\lim_{x \to a^+} f(x) = \infty$
- $\lim_{x \to a^-} f(x) = \infty$, $\lim_{x \to a^+} f(x) = \text{finite}$

- Example: $f(x) = \begin{cases} x^2, & x < 0 \\ 1/x, & x > 0 \end{cases}$



$\lim_{x \to 0^-} f(x) = 0$, $\lim_{x \to 0^+} = +\infty = \text{DNE}$

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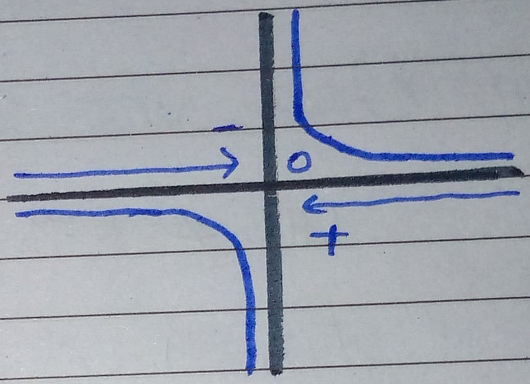
Date:

Case 2: One side goes to $+\infty$, other to $-\infty$

◆ Condition:

- $\lim_{x \to a^-} f(x) = -\infty$, $\lim_{x \to a^+} f(x) = +\infty$ (or vice versa)

- Example: $f(x) = \frac{1}{x}$



At $x = 0$: $\lim_{x \to 0^-} f(x) = -\infty$

$\lim_{x \to 0^+} f(x) = +\infty$

✗ Limit does not exist because both side diverge in opposite directions.

Case 3: Both sides go to $+\infty$ or $-\infty$

◆ Condition:

- $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = \pm\infty$

- Even though they agree, the limit still does not exist, because the value is infinite, not a finite number.

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When Does the Limit Exist?

Date:

$\lim_{x \to a} f(x)$ exists if and only if:

- $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

★ Both sides must approach the same value.

When Does the Limit Not Exist?

When, LHL ≠ RHL

We will study 4 cases where this happens.

Case 1: Left-hand limit ≠ Right-hand limit

- Condition:

$$\lim_{x \to -a} f(x) \neq \lim_{x \to a^+} f(x)$$

- Example:

$$f(x) = \begin{cases} 1, & x < 0 \\ 2, & x \geq 0 \end{cases}$$

- $\lim_{x \to 0^-} f(x) = 1$
- $\lim_{x \to 0^+} f(x) = 2$

✗ So, the limit does not exist at $x = 0$.

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Date:

Q $\lim_{x \to 0} \frac{1-\cos x}{x}$

$= \lim_{x \to 0} \frac{1-\cos x}{x} \times \frac{(1+\cos x)}{(1+\cos x)}$

$= \lim_{x \to 0} \frac{1-\cos^2 x}{x(1+\cos x)}$

$= \lim_{x \to 0} \frac{\sin^2 x}{x(1+\cos x)}$

$= \lim_{x \to 0} \frac{\sin x}{x} \cdot \frac{\sin x}{1+\cos x}$

$= \lim_{x \to 0} \frac{\sin x}{x} \cdot \lim_{x \to 0} \frac{\sin x}{1+\cos x}$

$= 1 \cdot \frac{0}{1} = 0$

Q. $\lim_{x \to 0} \frac{\sin(3x)}{x}$

$= \lim_{x \to 0} \frac{\sin 3x}{x}$

$= \lim_{x \to 0} \frac{\sin 3x}{3x} \cdot 3$

$= 1 \cdot 3 = 3$

Q. $\lim_{x \to 0} \frac{\tan(x)}{x}$

$= \lim_{x \to 0} \frac{\sin x / \cos x}{x}$

$= \lim_{x \to 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x}$

$= \lim_{x \to 0} \frac{\sin x}{x} \cdot \lim_{x \to 0} \frac{1}{\cos x}$

$= 1 \cdot \frac{1}{1} = 1$

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Trignometric Limits :- Date:

These limits usually involve $\sin(x)$, $\cos(x)$, or $\tan(x)$ as $x \to 0$, and they are very important to memorize and understand.

$$\therefore \lim_{x \to a} \sin(x) = \sin(a)$$

$$\therefore \lim_{x \to a} \cos(x) = \cos(a)$$

$$\therefore \lim_{x \to 0} \frac{\sin(x)}{x} = 1$$

$$\therefore \lim_{x \to 0} \frac{\tan(x)}{x} = 1$$

$$\therefore \lim_{x \to 0} \frac{1 - \cos x}{x} = 0$$

$$\lim_{x \to 0} \frac{1 - \cos(ax)}{x^2} = \frac{a^2}{2}$$

$$\lim_{x \to 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$$

Very Common Trick :

For :

$$\lim_{x \to 0} \frac{\sin(ax)}{bx}$$

Use :

$$\lim_{x \to 0} \frac{\sin(ax)}{ax} \cdot \frac{a}{b} = \frac{a}{b}$$

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Limit of a Piecewise Function :-

Date:

To find limit $\lim_{x \to a} f(x)$ where $f(x)$ is piecewise,

we must check : $\lim_{x \to a^-} f(x)$ (left-hand limit)

$\lim_{x \to a^+} f(x)$ (right-hand limit)

- If both limits exists and are equal, then the limit exists. Otherwise not.

Example 1: $f(x) = \begin{cases} x^2, & x < 1 \\ 2x+1, & x \geq 1 \end{cases}$, find $\lim_{x \to 1} f(x)$

Left-hand limit : $\lim_{x \to 1^-} x^2 = 1^2 = 1$

Right-hand limit : $\lim_{x \to 1^+} 2x+1 = (2(1)+1) = 3$

Since $1 \neq 3$, $\lim_{x \to 1} f(x)$ = does not exist.

Example 2: $f(x) = \begin{cases} x+2, & x < 3 \\ 5, & x = 3 \\ x^2-1, & x > 3 \end{cases}$, find $\lim_{x \to 3} f(x)$:

Left-hand limit : $\lim_{x \to 3^-} x+2 = 3+2 = 5$

Right hand limit : $\lim_{x \to 3^+} x^2-1 = 9-1 = 8$, $5 \neq 8 =$ DNE

* Even though $f(3) = 5$, that's irrelevant for the limit.

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1) What is Continuity :- Date:

* Intuitive Idea :-

A function is continuous at a point if you can draw its graph at that point without lifting your pen.
In simple words : No sudden jumps, holes, or breaks at that point.

Definition of Continuity at a Point :-

A function $f(x)$ is continuous at a point $x = a$ if the following 3 conditions are met :

1) $f(a)$ is defined

Means there is a real value for the function at $x = a$.

2) $\lim_{x \to a} f(x)$ exists

Means the left hand and right hand limits both exist and are equal.

3) $\lim_{x \to a} f(x) = f(a)$

Means the limit value matches the actual function value.

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Date:

Most famous use:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

We can't directly substitute and we don't yet have limit laws to ~~proof~~ prove it. So we use the sandwitch theorem.

From geometry:

$$\cos x \leq \frac{\sin x}{x} \leq 1 \quad \left(\text{when } 0 < x < \frac{\pi}{2}\right)$$

Now take limit as $x \to 0$:

$$\lim_{x \to 0} \cos x = 1 \quad \text{and} \quad \lim_{x \to 0} 1 = 1$$

So by Sandwitch Theorem:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \quad \text{(Proven)}.$$

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Sandwitch Theorem

Date: _____

What is it?

If a function $f(x)$ is "squeezed" between two other functions $g(x)$ and $h(x)$, and both of them have the same limit at a point, then the function $f(x)$ also has that same limit at that point.

The Theorem Says :-

If $$g(x) \leq f(x) \leq h(x)$$

for all x near a,

and

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} f(x) = L$$

Usefulness :-

When its hard to find the limit of a function directly, we "trap" it between two simpler functions.

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Date:

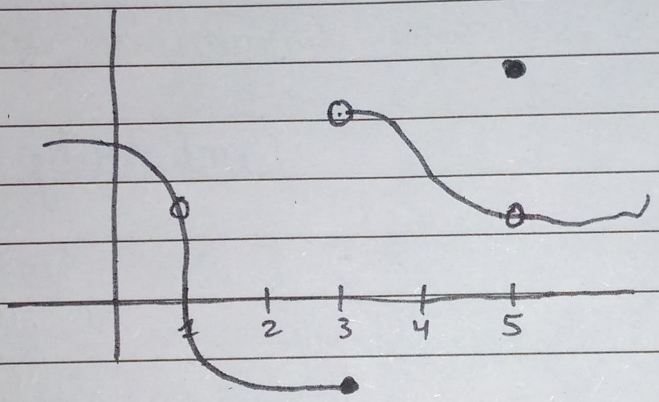
Example: $f(x) = x^2$, check continuity at $x = 2$.

1) $f(2) = 4$ ✓

2) $\lim_{x \to 2} x^2 = 4$ ✓

3) limit = Function Value ✓

So, $f(x) = x^2$ is continuous at $x = 2$



at $x = 1$,
$f(x)$ is discontinue at 1, because the function is not defined. (not continuous)

at $x = 3$,
function is defined but limit does not exist. (✓)

at $x = 5$,
function is defined and limit also exist but both are not same. (not continuous).

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Continuity in Domain- Shortcut Rule & Special Cases :-

Date:

* Golden Shortcut Rule :-

If a function is defined at a point and belongs to a type of function that is always continuous in its domain, then it is automatically continuous at the point.

This means: You don't need to check the 3 conditions if the point is inside the domain and the function type is "nice 😊".

Functions that are always continuous on their domains.

Function Type	Continuity
Constant $f(x) = c$	Continuous Everywhere
Identity $f(x) = x$	Continuous Everywhere
Polynomial	Continuous Everywhere
Rational $p(x)/q(x)$	Continuous where $q(x) \neq 0$
Squar root $\sqrt{x}$	Continuous where $x \geq 0$
Exponential a^x	Continuous Everywhere
Logarithmic $\log x$	Continuous for $x > 0$
Trigonometric (sin, cos)	Continuous Everywhere
(tan, cot, sec, cosec)	Continuous where defined

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Important Notes on Rational Functions :-

Date:

- Rational functions are only continuous at values where the denominator $\neq 0$.
- If $x = a$ makes denominator 0, then :
  - $f(a)$ is undefined
  - → Not continuous at $x = a$.

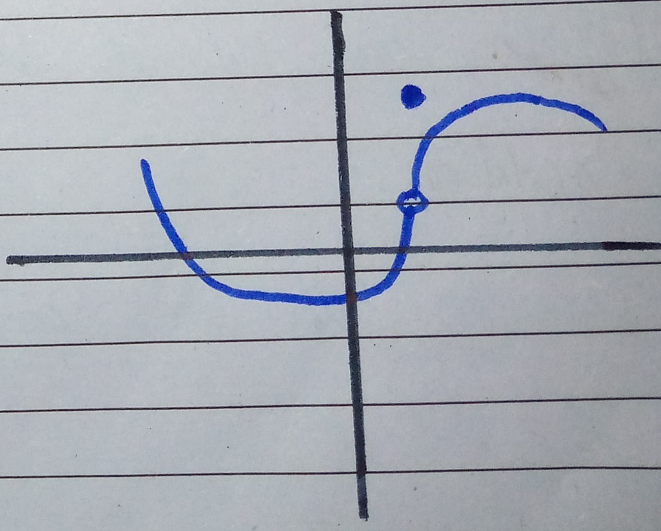
* But if $x = a \in$ domain, then the function is automatically continuous at $x = a$.

2) Types of Discontinuities :-

There are 3 major Types you must know :

1) Removeable Discontinuity (hole) :-

The graph has a hole at that point, but the rest of the graph is smooth.



◆ Condition :

- Limits exist.
- Function is either undefined or not equal to the limit at that point.

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2) Jump Discontinuity :-

Date:

Graph jumps suddenly from one point to another — like stepping up a stair.

- Condition:

- Left-hand limit ≠ Right-hand limit
- So, overal limit does not exist.

Example:

$$f(x) = \begin{cases} 1 & x < 2 \\ 3 & x \geq 2 \end{cases}$$

Here, $\lim_{x \to 2^-} = 1$, $\lim_{x \to 2^+} = 3$ → not equal.

* Jump Discontinuity at $x = 2$.

3) Infinite Discontinuity :-

Graph shoots up or down to infinity — vertical asymptote

- Condition:

- One or both one side limits are ∞ or $-\infty$.

SP PAPER PRODUCT

Intermediate Value Theorem (IVT)

(Also called The Intermediate Theorem) :-

Date: _____